



W21

प्रश्नपुस्तिका क्रमांक
BOOKLET NO.

145413

संच क्र.

A

प्रश्नपुस्तिका

विहित अभ्यासक्रमावर आधारित संबंधित घटक

एकूण प्रश्न : 100

एकूण गुण : 200

वेळ : 1 (एक) तास

सूचना

- (1) सदर प्रश्नपुस्तिकेत 100 अनिवार्य प्रश्न आहेत. उमेदवारांनी प्रश्नांची उत्तरे लिहिण्यास सुरुवात करण्यापूर्वी या प्रश्नपुस्तिकेत सर्व प्रश्न आहेत किंवा नाहीत याची खात्री करून घ्यावी. तसेच अन्य काही दोष आढळल्यास ही प्रश्नपुस्तिका समवेक्षकांकडून लगेच बदलून घ्यावी.
- (2) आपला परीक्षा-क्रमांक ह्या चौकोनात न विसरता बॉलपेनने लिहावा.
- (3) वर छापलेला प्रश्नपुस्तिका क्रमांक तुमच्या उत्तरपत्रिकेवर विशिष्ट जागी उत्तरपत्रिकेवरील सूचनेप्रमाणे न विसरता नमूद करावा.
- (4) या प्रश्नपुस्तिकेतील प्रत्येक प्रश्नाला 4 पर्यायी उत्तरे सुचविली असून त्यांना 1, 2, 3 आणि 4 असे क्रमांक दिलेले आहेत. त्या चार उत्तरांपैकी सर्वात योग्य उत्तराचा क्रमांक उत्तरपत्रिकेवरील सूचनेप्रमाणे तुमच्या उत्तरपत्रिकेवर नमूद करावा. अशा प्रकारे उत्तरपत्रिकेवर उत्तर-क्रमांक नमूद करताना तो संबंधित प्रश्न-क्रमांकासमोर छायांकित करून दर्शविला जाईल याची काळजी घ्यावी. ह्याकरिता फक्त काळ्या शाईचे बॉलपेन वापरावे, पेन्सिल वा शाईचे पेन वापरू नये.
- (5) सर्व प्रश्नांना समान गुण आहेत. यास्तव सर्व प्रश्नांची उत्तरे द्यावीत. घाईमुळे चुका होणार नाहीत याची दक्षता घेऊनच शक्य तितक्या वेगाने प्रश्न सोडवावेत. क्रमाने प्रश्न सोडविणे श्रेयस्कर आहे पण एखादा प्रश्न कठीण वाटल्यास त्यावर वेळ न घालविता पुढील प्रश्नाकडे वळावे. अशा प्रकारे शेवटच्या प्रश्नापर्यंत पोहोचल्यानंतर वेळ शिल्लक राहिल्यास कठीण म्हणून वगळलेल्या प्रश्नांकडे परतणे सोईस्कर ठरेल.
- (6) उत्तरपत्रिकेत एकदा नमूद केलेले उत्तर खोडता येणार नाही. नमूद केलेले उत्तर खोडून नव्याने उत्तर दिल्यास ते तपासले जाणार नाही. एकापेक्षा जास्त उत्तरे नमूद केल्यास ते उत्तर चुकीचे धरले जाईल व त्या चुकीच्या उत्तराचे गुण वजा केले जातील.
- (7) प्रस्तुत परीक्षेच्या उत्तरपत्रिकांचे मूल्यांकन करताना उमेदवाराच्या उत्तरपत्रिकेतील योग्य उत्तरांचा गुण दिले जातील. तसेच “उमेदवाराने वस्तुनिष्ठ बहुपर्यायी स्वरूपाच्या प्रश्नांची दिलेल्या चार उत्तरांपैकी सर्वात योग्य उत्तरेच उत्तरपत्रिकेत नमूद करावीत. अन्यथा त्यांच्या उत्तरपत्रिकेत सोडविलेल्या प्रत्येक चुकीच्या उत्तराकरिता 25% किंवा 1/4 एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.”

परीक्षा-क्रमांक									

केंद्राची संकेताक्षरे

शेवटचा अंक

ताकीद

ह्या प्रश्नपुस्तिकेसाठी आयोगाने विहित केलेली वेळ संपेपर्यंत ही प्रश्नपुस्तिका आयोगाची मालमत्ता असून ती परीक्षा कक्षात उमेदवाराला परीक्षेसाठी वापरण्यास देण्यात येत आहे. स्पर्धा परीक्षा घेण्यासाठी, प्राधिकृत केलेली, नेमलेली किंवा कर्तव्ये सोपविलेली कोणतीही व्यक्ती, प्रश्नपुस्तिका उघडण्यासाठी व वितरित करण्यासाठी निश्चित केलेल्या वेळेपूर्वी अशी प्रश्नपुस्तिका किंवा तिचा कोणताही भाग किंवा तिची प्रत उघडणार नाही, फोडणार नाही किंवा ताब्यात घेणार नाही किंवा पाहणार नाही किंवा सोडविणार नाही किंवा सोडविण्यास सहाय्य करणार नाही; असे केल्यास “महाराष्ट्र स्पर्धा परीक्षा (अनुचित मार्गास प्रतिबंध) अधिनियम - 2024” मधील तरतुदीनुसार सदर व्यक्ती तीन वर्षांपेक्षा कमी नसेल परंतु, जी पाच वर्षांपर्यंत वाढविता येईल इतक्या मुदतीची कारावासाची शिक्षा आणि दहा लाख रुपयांपर्यंतच्या द्रव्यदंडास पात्र होईल. तसेच ह्या प्रश्नपुस्तिकेसाठी आयोगाने विहित केलेली वेळ संपण्याआधी ही प्रश्नपुस्तिका अनधिकृतपणे बाळगणे हा सुद्धा गुन्हा असून तसे करणारी व्यक्ती आयोगाच्या कर्मचारीवृंदापैकी, तसेच परीक्षेच्या पर्यवेक्षकीयवृंदापैकी असली तरीही अशा व्यक्तीविरुद्ध उक्त अधिनियमानुसार कारवाई करण्यात येईल व दोषी व्यक्ती शिक्षेस पात्र होईल.

पुढील सूचना प्रश्नपुस्तिकेच्या अंतिम पृष्ठावर पहा.

पर्यवेक्षकांच्या सूचनेविना हे सील उघडू नये

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कच्च्या कामासाठी जागा / SPACE FOR ROUGH WORK



1. Which of the following is natural anticoagulant ?

- | | |
|--------------|-------------------|
| (1) Warfarin | (2) Acenocoumarol |
| (3) Heparin | (4) Disodium EDTA |

2. Which of the following statements adhere to the standards set by the Indian Pharmacopoeia 2010 for drawing the whole human blood ?

- A. Appropriate antimicrobial preservative is added to avoid microbial contamination.
- B. The blood drawn aseptically through a closed system in to a suitable sterile container.
- C. A specific amount of anticoagulant EDTA must be added in each before collection of blood.
- D. With every container of blood at least one separate sample mixed with appropriate quantity of anticoagulant is collected for compatibility and other tests.

The most appropriate statements are _____.

- | | | | |
|-------------|-------------|-------------|-------------|
| (1) A and B | (2) B and D | (3) C and D | (4) A and D |
|-------------|-------------|-------------|-------------|

3. Deficiency of _____ prevents normal blood coagulation.

- | | |
|---------------|---------------|
| (1) Vitamin A | (2) Vitamin C |
| (3) Vitamin D | (4) Vitamin K |

4. Normal Lymphocytes counts in adult blood is _____.

- | | |
|--|--|
| (1) $(0.5 \text{ to } 1.5) \times 10^9/\text{l}$ | (2) $(1.5 \text{ to } 2.5) \times 10^9/\text{l}$ |
| (3) $(1.5 \text{ to } 3.5) \times 10^9/\text{l}$ | (4) $(3.5 \text{ to } 5.5) \times 10^9/\text{l}$ |

5. Suppression of cell-mediated immunity in HIV infection is due to _____.

- (1) Excess of CD4 and deficiency of T cells
- (2) Deficiency of CD4 and excess of T cells
- (3) Deficiency of CD4 and deficiency of T cells
- (4) Excess of CD4 and excess of T cells



6. Which hypersensitivity reaction, based on its immune mechanism, is also known as Atopic hypersensitivity ?

- | | |
|-------------------------------|------------------------------|
| (1) Type I hypersensitivity | (2) Type II hypersensitivity |
| (3) Type III hypersensitivity | (4) Type IV hypersensitivity |

7. Select the most appropriate statements relevant for HBsAg.

- A. This antigen was termed Australia antigen.
- B. It is derived from core protein and present in the nuclei of hepatocytes.
- C. It appears in the blood after around six weeks of infection.
- D. This antigen largely contributes to pathogenicity of virus.

The most appropriate statements are _____.

- | | | | |
|-------------|-------------|-------------|-------------|
| (1) A and B | (2) B and D | (3) C and D | (4) A and C |
|-------------|-------------|-------------|-------------|

8. Selectivity of calcium channel blockers for cardiovascular action depends on following aspects :

- a. Dihydropyridines have less effect on vascular smooth muscle as compared to cardiac effect.
- b. All dihydropyridines have the common mechanism of action.
- c. Nicardipine has selective action on the coronary blood vessels.
- d. Nimodipine has selective on cerebral blood vessels.
- e. Amlodipine produces no tachycardia.
- f. Diltiazem and verapamil are more selective for cardiac muscle.

- (1) All aspects are true
- (2) Aspect no. 'a' and 'e' are false and all other aspects are true
- (3) All aspects are true except aspects no. 'd'
- (4) Aspect no. 'b', 'd', 'e' and 'f' are true and all other aspects are false

कच्चा कामासाठी जागा / SPACE FOR ROUGH WORK

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9. Which type of toxicity is associated with ion tapping effects of Aspirin ?

- | | |
|----------------------|----------------------|
| (1) Hepato toxicity | (2) Gastric toxicity |
| (3) Dermato toxicity | (4) Musculo toxicity |

10. Which one of these is not insulin secretagogue ?

- | | |
|---------------|-----------------|
| (1) Glyburide | (2) Nateglinide |
| (3) Miglitol | (4) Repaglinide |

11. Which one of these drugs, do not inhibit tubulin and reduce helminth infestation ?

- | | |
|-----------------|---------------------|
| (1) Albendazole | (2) Metronidazole |
| (3) Mebendazole | (4) Triclabendazole |

12. Imatinib is exhibiting its action by inhibition of _____.

- | |
|--|
| (1) Protein kinase activities of Jak ₁ and Jak ₂ |
| (2) Cyclin dependant kinase 3 (CDK) |
| (3) BCR-ABL kinase |
| (4) Serine / threonine protein kinase |

13. Which of the following features in rifampicin is necessary for enzyme binding ?

- | |
|---|
| (1) Double bonds in macrocyclic ring |
| (2) Free hydroxy group at C-1, C-8, C-21 and C-23 |
| (3) Macrocyclic ring |
| (4) Lactone ring group |

14. Aliskiren acts by _____.

- | |
|--|
| (1) Inhibiting angiotensin converting enzyme |
| (2) Blocking AT angiotensin receptors |
| (3) Inhibiting enzyme activity of renin |
| (4) Opening K ⁺ channels |



15. Eszopiclone is a short acting hypnotic drug use to induce sleep, acts at which of the following receptor ?

- (1) Benzodiazepine site on GABA – A receptor containing beta-1 sub unit
- (2) Benzodiazepine site on GABA – A receptor containing alfa-2 sub unit
- (3) Benzodiazepine site on GABA – A receptor containing alfa-1 sub unit
- (4) Benzodiazepine site on GABA – A receptor containing beta-2 sub unit

16. Verapamil is a _____ antiarrhythmic drug act by blocking voltage sensitive calcium channel.

- (1) Class I
- (2) Class II
- (3) Class III
- (4) Class IV

17. Which of the following drugs is a cardioselective β -blocker ?

- (1) Sotalol
- (2) Propranolol
- (3) Labetalol
- (4) Esmolol

18. In which of the following drug discovery step, pharmacokinetic and safety properties (ADMET) are assessed ?

- (1) Lead discovery
- (2) Lead optimization
- (3) Target validation
- (4) Hit identification

19. Flucytosine acts by _____.

- (1) Disrupting microtubule function
- (2) Disrupting the synthesis of both RNA and DNA
- (3) Inhibiting ergosterol synthesis
- (4) Inhibiting fungal cell wall synthesis

कच्च्या कामासाठी जागा / SPACE FOR ROUGH WORK

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20. Zinc binding group/moiety in ACE inhibitor is _____.

- | | |
|----------------|-----------|
| (1) Alkyl | (2) Amide |
| (3) Sulfhydryl | (4) Ester |

21. Toxins bound tightly to protein or which are highly lipid-soluble are removed by _____.

- | | |
|-------------------|-----------------------------|
| (1) Dialysis | (2) Forced diuresis |
| (3) Hemoperfusion | (4) Charcoal administration |

22. What is mechanism of Thalidomide teratogenicity ?

- (1) Angiogenesis
- (2) DNA disruption
- (3) Interference with cellular signaling
- (4) Enzyme inhibition

23. Antidote of methanol poison is _____.

- | | |
|-------------|---------------------------|
| (1) Ethanol | (2) Glucagon |
| (3) Oxygen | (4) Pralidoxime (2 – PAM) |

24. Penicillamine is not a preferred antidote for _____ poisoning.

- | | |
|------------|-------------|
| (1) Lead | (2) Mercury |
| (3) Copper | (4) Iron |

25. The chlorinated hydrocarbons used as insecticides, acts at the picrotoxin binding site of _____ receptors in the insects.

- | | |
|--------------|--------------|
| (1) D – 1 | (2) D – 2 |
| (3) GABA – A | (4) GABA – B |



26. Hybridoma cell technology is most widely employed for the large-scale production of which of the following ?

- | | |
|---------------------------|-------------------------------|
| (1) Polyclonal antisera | (2) Monoclonal antibodies |
| (3) Recombinant cytokines | (4) Attenuated viral vaccines |

27. The dioctyl phthalate (DOP) test is primarily employed as a validation method for assessing the efficiency of which of the following system ?

- (1) HEPA filter
- (2) Autoclave
- (3) UV radiation
- (4) Ethylene oxide gas sterilizers

28. Mac Conkey's medium is selective and differential media for _____.

- (1) Escherichia coli
- (2) Helicobacter pylori
- (3) Staphylococcus aureus
- (4) Bacillus subtilis

29. An intact HEPA filter should be capable of removing at least _____ of particulates of 0.3 micrometer and larger in diameter.

- | | | | |
|---------|---------|------------|-----------|
| (1) 97% | (2) 99% | (3) 99.97% | (4) 99.7% |
|---------|---------|------------|-----------|

30. Ebola virus causing human hemorrhagic fever belongs to family _____.

- | | |
|------------------|-------------------|
| (1) Filoviridae | (2) Coronaviridae |
| (3) Retroviridae | (4) Reoviridae |

31. The holding time in autoclave for sterilization of articles kept at 134°C and 30 psi pressure is _____.

- | | |
|----------------|----------------|
| (1) 30 minutes | (2) 15 minutes |
| (3) 10 minutes | (4) 3 minutes |

कच्चा कामासाठी जागा / SPACE FOR ROUGH WORK

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32. Which micro-organism is routinely employed for microbiological assay of Vitamin B12 ?

- | | |
|-------------------------------|-----------------------|
| (1) Lactobacillus leichmannii | (2) Escherichia coli |
| (3) Saccharomyces cerevisiae | (4) Bacillus subtilis |

33. Which condition is considered the most severe and life-threatening form of pneumococcal infection ?

- | | |
|------------------------|---------------------|
| (1) Acute otitis media | (2) Lobar pneumonia |
| (3) Meningitis | (4) Sinusitis |

34. All micro-organisms are removed or killed on or in item is called _____.

- | | |
|-------------------|---------------------|
| (1) Disinfection | (2) Sanitization |
| (3) Sterilization | (4) Decontamination |

35. Match the following pharmaceutical products produced by micro-organisms.

- | | |
|-------------------|----------------------|
| a. Riboflavin | m. Propionibacterium |
| b. Amino acids | n. Corynebacterium |
| c. Citric acid | o. Aspergillus |
| d. Cyanocobalamin | p. Eremothecium |

- (1) a – m, b – o, c – n, d – p
(2) a – n, b – p, c – m, d – o
(3) a – o, b – m, c – p, d – n
(4) a – p, b – n, c – o, d – m

36. Hoogsteen base pairing allows formation of which DNA structure ?

- | | |
|-----------------|--------------------|
| (1) A – DNA | (2) Z – DNA |
| (3) Triplex DNA | (4) Quadruplex DNA |

कच्च्या कामासाठी जागा / SPACE FOR ROUGH WORK

P.T.O.

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37. Galactolipids, localized in the internal membranes of chloroplasts, make up _____ of the total membrane lipids of a vascular plant.

- | | |
|----------------|----------------|
| (1) 70% to 80% | (2) 50% to 60% |
| (3) 60% to 70% | (4) 80% to 90% |

38. Which option is true for reverse transcriptase ?

- (1) It is found in retroviruses
- (2) It synthesizes DNA from RNA
- (3) It requires primer for synthesis
- (4) All of the above

39. The conserved HIV epitopes used in chimeric vaccines are typically _____.

- (1) Poorly immunogenic without a carrier
- (2) Highly immunogenic on their own
- (3) Toxic to host cells
- (4) Unrelated to viral proteins

40. Name the pathway for glucose synthesis by non-carbohydrate precursors.

- | | |
|---------------------|--------------------|
| (1) Glycogenesis | (2) Glycolysis |
| (3) Gluconeogenesis | (4) Glycogenolysis |

41. Identify correct statement from the following.

- a. Hydrocortisone and cortisone belong to glucocorticoids.
 - b. Aldosterone belongs to mineralocorticoid.
 - c. 11-deoxycorticosterone is an example of mineralocorticoid.
- (1) Only a is correct statement
 - (2) b and c are correct statements
 - (3) a and b are correct statements
 - (4) a, b, c all are correct statements

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42. Identify true/false statement from the following.
- Steroid hormones in mammals are biosynthesized from cholesterol, via the mevalonate pathway.
 - Steroid hormones in mammals are biosynthesized from cholesterol, via the shikimic acid pathway.
 - Conversion of cholesterol to pregnenolone is the rate limiting step in steroid hormone biosynthesis.
- a and c are true while b is false
 - b and c are true while a is false
 - a and b are true while c is false
 - only a is correct
-

43. Why are aryl diazonium salts prepared at $0 - 5^{\circ}\text{C}$?
- To increase electrophilicity
 - To prevent oxidation of amines
 - To prevent decomposition of diazonium salts
 - To increase the reaction rates
-

44. The no. of optical isomers possible for 2-Bromo-3-chlorobutane are _____.
- (1) 8 (2) 10 (3) 4 (4) 2
-

45. Axillary (underarm) odour is primarily produced due to _____.
- Excess secretion from eccrine glands
 - Action of bacteria on nutrient present in apocrine secretion
 - Oxidation of sweat by air
 - Friction between skin surfaces
-

46. The rate controlling membrane used in Pilo 20 Ocuser is _____.
- Polyvinyl acetate
 - Ethylene vinyl acetate
 - Polyethylene glycol
 - Poly (lactic-co-glycolic acid)
-



47. An excipient used to promote flow of tablet granules or powder materials by reducing friction between particles is _____.

- | | |
|---------------|------------------|
| (1) Lubricant | (2) Antiadherent |
| (3) Glidant | (4) Diluent |

48. The Investigational New Drug application is part of _____.

- | | |
|---------------------|---------------------|
| (1) 21 CFR Part 312 | (2) 21 CFR Part 316 |
| (3) 21 CFR Part 111 | (4) 21 CFR Part 214 |

49. Droplet size of monophasic microemulsion is _____.

- | | |
|-----------------|-----------------|
| (1) 10 – 100 nm | (2) 20 – 200 nm |
| (3) 30 – 300 nm | (4) 50 – 500 nm |

50. Preformulation studies of newly synthesized drug focus on which of the following parameters ?

- (1) Physicochemical properties of new drug
- (2) Regulatory composition of new drug
- (3) Patent filing of new drug
- (4) Animal studies of new drug

51. The segregation of tablet into two or more distinct layers is termed as _____.

- | | |
|--------------|----------------|
| (1) Capping | (2) Lamination |
| (3) Chipping | (4) Cracking |

कच्च्या कामासाठी जागा / SPACE FOR ROUGH WORK

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52. Which suppository formulation has the rapid release rate as it relate to the drug and base characteristics ?

- (1) Oil-soluble drug : oily base
- (2) Water-soluble drug : oily base
- (3) Oil-soluble drug : water miscible base
- (4) Water-insoluble drug : oily base

53. Which temperature range is generally specified for refrigerated pharmaceutical raw materials ?

- (1) 8°C to 15°C
- (2) 2°C to 8°C
- (3) 15°C to 25°C
- (4) - 20°C to - 10°C

54. Onset time for systemic routes of administration of suppositories, ointments and creams through rectal route is _____.

- (1) 5 – 10 minutes
- (2) 15 – 30 minutes
- (3) 30 – 60 minutes
- (4) 60 – 90 minutes

55. Chemical name of Tween 80 is _____.

- (1) Polyoxyethylene sorbitan monostearate
- (2) Polyoxyethylene sorbitan monooleate
- (3) Polyoxyethylene sorbitan monolaurate
- (4) Sorbitan monolaurate

56. For oral solid dosage forms, the pilot batch size should be generally _____ % of production scale.

- (1) 5
- (2) 10
- (3) 25
- (4) 30

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57. BET nitrogen adsorption method is used to measure _____.

- | | |
|----------------------------|---------------------------|
| (1) Hardness | (2) Flowability of powder |
| (3) Surface area of powder | (4) Bulk density |

58. Shikonin, the enantiomer of alkannin obtained from _____ root and is used for herbal lipstick production.

- | | |
|-----------------------------|--------------------------------|
| (1) Lithospermum officinale | (2) Lithospermum erythrorhizon |
| (3) Lithospermum canescens | (4) Lithospermum incisum |

59. To overcome problem of hygroscopicity of herbal extracts, the following excipient is widely used _____.

- | | | | |
|------------|----------|-------------|----------------|
| (1) Kaolin | (2) Talc | (3) Alumina | (4) Silica gel |
|------------|----------|-------------|----------------|

60. Concentrated sugar solutions are quite resistant to microbial growth because _____.

- | |
|---|
| (1) More availability of the water required for growth of micro-organisms |
| (2) Unavailability of the water required for the growth of micro-organisms |
| (3) An efficient nutrient medium required for the growth of micro-organisms |
| (4) Not sufficient nutrient medium required for the growth of micro-organisms |

61. In Bragg's law for XRD, the angle between the diffracted beam and transmitted beam is always _____.

- | | | | |
|--------------|---------------|---------------|---------------|
| (1) θ | (2) 2θ | (3) 3θ | (4) 4θ |
|--------------|---------------|---------------|---------------|

62. The cause and effect diagram, widely applied in TQM was introduced by which quality expert ?

- | | |
|--------------------|---------------------|
| (1) Edward Deming | (2) Joseph M. Juran |
| (3) Phillip Crosby | (4) Kaoru Ishikawa |

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63. Chemical shift is measured in _____.

- (1) Hz (2) ppm (3) Tesla (4) Joules
-

64. The calibration wavelength scale of a UV/visible spectrophotometer is checked by determining the specified wavelength maxima of a _____ solution of holmium perchlorate.

- (1) 1% w/v (2) 2% w/v (3) 5% w/v (4) 10% w/v
-

65. The chromophore is defined as _____.

- (1) A group that decreases absorption of electromagnetic radiation
(2) A group for characteristic absorption of electromagnetic radiation
(3) A group that shifts the absorption maximum to shorter wavelength
(4) A group unable to absorb electromagnetic radiation
-

66. The primary function of the carrier gas in GLC is to _____.

- a. React with the analytes for detection.
b. Maintain a high pressure and deliver the sample through a column at a reproducible rate.
c. Maintain a high pressure and deliver the sample through a stationary liquid phase at a reproducible rate.
d. Maintain a high thermal insulation for the column.

- (1) Only a, b and c (2) Only b, c and d
(3) Only b (4) Only c
-

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67. X-Ray diffraction technique is based on _____.

- | | |
|-------------------|------------------------|
| (1) Bragg's law | (2) Beer Lambert's law |
| (3) Pasteur's law | (4) Fick's law |

68. In UV spectroscopy, which of the following electronic transitions requires the highest energy ?

- | | | | |
|-----------------------------------|-----------------------------|---------------------------|------------------------------|
| (1) $\sigma \rightarrow \sigma^*$ | (2) $\pi \rightarrow \pi^*$ | (3) $n \rightarrow \pi^*$ | (4) $n \rightarrow \sigma^*$ |
|-----------------------------------|-----------------------------|---------------------------|------------------------------|

69. Near IR region is _____.

- | | |
|--|---|
| (1) 0.8 μm to 2.5 μm | (2) 2.5 μm to 50 μm |
| (3) 50 μm to 1000 μm | (4) 10 μm to 100 μm |

70. In AAS, liquid samples that are too dilute should be _____.

- (1) Discarded and remade
- (2) Evaporated or concentrated by solvent extraction
- (3) Analyzed without any further treatment
- (4) Mixed with strong acid

71. All of the following States are major pioneers of pooled procurement of essential medicines in India, Except _____.

- | | |
|----------------|-------------|
| (1) Tamil Nadu | (2) Kerala |
| (3) Delhi | (4) Gujarat |

72. The validity of retail drug store licence is _____.

- | | |
|-------------|-------------|
| (1) 1 year | (2) 3 years |
| (3) 5 years | (4) 8 years |

73. When the telephonic orders for controlled drugs are accepted, doctors must sign their initials within _____.

- | | |
|------------|------------|
| (1) 12 hrs | (2) 24 hrs |
| (3) 6 hrs | (4) 8 hrs |

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74. Science of collecting, monitoring, researching, assessing and evaluating information from healthcare providers and patients on the adverse effects of medications is known as _____.

- | | |
|-------------------------|-----------------------|
| (1) Pharmacovigilance | (2) Clinical trials |
| (3) Observational study | (4) Qualitative study |

75. The “Envelope System” in hospital pharmacy is primarily used for _____.

- (1) Dispensing controlled substances
- (2) Charging non-floor stock medications
- (3) Inventory of narcotics
- (4) Sterile compounding

76. Which of the following is/are included in Schedule K of the Drugs and Cosmetics Rules ?

- a. Nicotine gum.
 - b. Oral Rehydration Salts.
 - c. Analgesic balms.
 - d. Prescription Drugs for treatment of hypertension.
- | | |
|---------------------|------------------|
| (1) Only c | (2) Only a and c |
| (3) Only a, b and c | (4) Only c and d |

77. According to Cosmetic Rules, 2020 (under Section 18(3)), which substance is prohibited in imported cosmetics ?

- | | |
|----------------------|---------------------|
| (1) Chlorhexidine | (2) Hexachlorophene |
| (3) Benzoyl peroxide | (4) Sorbic acid |

78. The latest overhaul of Schedule M has some provisions like _____.

- a. Specific requirements for biological products.
 - b. Quality risk management.
 - c. Periodic review of products meant for export.
 - d. Data integrity of computerised systems.
- | | |
|------------------|---------------------|
| (1) Only c and d | (2) Only a, b and d |
| (3) Only c | (4) a, b, c and d |



79. The functions of Indian Pharmacopoeia Commission (IPC) include _____.

- a. Revision of the Indian Pharmacopoeia.
- b. Analysis of Drug samples sent by Drug Inspector.
- c. Acts as appellate analytical laboratory for drugs.
- d. Provides IP reference standards.

- | | |
|------------------|---------------------|
| (1) Only a | (2) Only a, c and d |
| (3) Only b and d | (4) Only a and d |

80. According to Medical Devices Rules, 2017, which of the following will be classified under Class D ?

- a. Pacemaker for heart
- b. Implantable stents
- c. Breast implant
- d. Reusable invasive surgical instrument

- | | |
|-------------------|-------------|
| (1) a, b, c and d | (2) a and b |
| (3) a, b and c | (4) only b |

81. The Drugs Technical Advisory Board (DTAB) constituted by the Central Government includes which of the following as ex-officio members ?

- a. Drugs Controller, India.
- b. Minister of Health and Family Welfare, Government of India.
- c. Director, Indian Veterinary Research Institute, Izatnagar.
- d. President, Pharmacy Council of India.

- | | |
|---------------------|------------------|
| (1) Only a, c and d | (2) Only a and d |
| (3) Only a, b and c | (4) Only a and b |

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82. Which ISO standard specifies the biocompatibility and animal performance study for investigational medical device ?

- | | |
|---------------|---------------|
| (1) ISO 9001 | (2) ISO 14001 |
| (3) ISO 13485 | (4) ISO 10993 |

83. If the cosmetic product is labelled with a false manufacture name and address, it will be classified as _____ as per D and C Act, 1940.

- | | |
|-----------------|---|
| (1) Misbranded | (2) Spurious |
| (3) Adulterated | (4) Cosmetics 'not of standard quality' |

84. Drugs may be imported into India by rail through Mohiassan (Mahisasan) Railway Station, which connects India with _____.

- | | |
|----------------|------------|
| (1) Pakistan | (2) Nepal |
| (3) Bangladesh | (4) Bhutan |

85. As per Standards of Cosmetics in Cosmetic Rules 2020, which of the following statements are correct ?

- A. Cosmetics intended for use in the area of eye should not contain mercury.
- B. In finished product of cosmetics, unintentional mercury should not exceed 1 ppm.
- C. Cosmetics should not contain more than 2 ppm of arsenic.
- D. In the soaps not to be used on babies, hexachlorophene can be used in concentration not exceeding 1% w/w.

- | | |
|----------------|-------------------|
| (1) A, B and C | (2) B, C and D |
| (3) A, B and D | (4) A, B, C and D |



86. Type D adverse drug reactions are characterized by which of the following ?

- (1) Dose related predictable effects, such as hypoglycemia with insulin
- (2) Diverse idiosyncratic unpredictable reactions, such as aplastic anemia
- (3) Delayed dose – independent effects, such as carcinogenicity and teratogenicity
- (4) Direct immediate hypersensitivity reactions, such as anaphylaxis

87. Hydroxykynurenic acid is present in _____.

- | | |
|------------------------|---------------------|
| (1) Ginkgo biloba | (2) Ocimum sanctum |
| (3) Terminalia chebula | (4) Bacopa monnieri |

88. Match the terms with descriptions.

Term	Description
a. Health supplement	i. Contains non-viable food components as added ingredients for modulation of gut microbiota
b. Nutraceutical	ii. Consists of a concentrated source of nutrients and/or other ingredients to supplement normal diet
c. Prebiotic food	iii. Contains live micro-organisms beneficial to human health
d. Probiotic food	iv. Consists of food with extracts, isolates and purified chemical compounds to help to maintain health

	a	b	c	d
(1)	ii	iv	iii	i
(2)	iv	ii	i	iii
(3)	ii	iv	i	iii
(4)	iii	iv	i	ii

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89. As per the DPCO formula, the ceiling price of a scheduled formulation is calculated using $P(C) = P(S) \times (1 + M/100)$.

Here, M, the margin to retailer, is equal to _____.

- (1) 10% (2) 14% (3) 16% (4) 20%

90. Key features of the Integrated Ayushman Bharat Pradhan Mantri Jan Arogya Yojana – Mahatma Jyotirao Phule Jan Arogya Yojana include _____.

- a. Universal Health Coverage for all citizens of Maharashtra
- b. Does not cover pre-existing diseases
- c. Scheme is fully paperless
- d. Health coverage of Rs. 5 lakhs per year per family

- (1) Only b, c and d (2) Only a, c and d
(3) a, b, c and d (4) Only a and d

91. Pradhan Mantri Jan Arogya Yojana _____.

- A. is universal health insurance
- B. includes emergency and critical care surgeries
- C. is applicable to people aged 60 and above only
- D. implements Jan Aushadhi Yojana

- (1) Only A and B (2) Only A and C
(3) Only B and C (4) Only A and D

92. As per World Health Organization, which of the following has emerged and considered as the top-ranked multiple drug-resistant fungi ?

- (1) Cryptococcus neoformans (2) Candida tropicalis
(3) Candida parapsilosis (4) Cryptococcus gattii



93. For original literary, dramatic, musical and artistic works, the term of copyright in India is for _____.

- (1) 50 years from the date of publication
- (2) Author's life plus 60 years
- (3) 60 years from the date of registration
- (4) Author's life plus 50 years

94. Which of the following Fixed Dose Combinations (FDCs) announced as irrational by Government of India and therefore, banned ?

- a. Doxycycline + Ornidazole + Bromelain + Lactobacillus rhamnosus + Lactobacillus reuteri RC
- b. Etodolac + Paracetamol + Serratiopeptidase
- c. Glimepiride 1 mg/2 mg + Metformin 500 mg SR tablet
- d. Tramadol HCl + Dicyclomine HCl + Domperidone

- (1) a, b, c and d
- (2) Only b and d
- (3) Only a, b and d
- (4) Only a

95. According to the definition given by Food Safety and Standards Act, Section 22; 'nutraceutical' includes _____.

- a. Ayurvedic/Siddha/Unani drug
- b. Plants or botanicals in the form of powder
- c. Proteins or amino acids
- d. Substances from animal origin

- (1) a, b, c and d
- (2) Only b, c and d
- (3) Only b and c
- (4) Only a, b and c

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96. B and E are good in Dramatics and Computer Science. A and B are good in Computer Science and Physics. A, D and C are good in Physics and History. C and A are good in Physics and Mathematics. D and E are good in History and Dramatics. Then, who is good in Physics, History and Mathematics but not in Computer Science ?
- (1) A (2) B (3) C (4) E
-
97. What number will be in the blank space 7, _____, 21, 31, 43 ?
- (1) 11 (2) 12 (3) 13 (4) 14
-
98. If 'MACHINE' is coded as 19 – 7 – 9 – 14 – 15 – 20 – 11, how will you code 'DANGER' ?
- (1) 11 – 7 – 20 – 16 – 11 – 24
(2) 13 – 7 – 20 – 9 – 11 – 25
(3) 10 – 7 – 20 – 13 – 11 – 24
(4) 13 – 7 – 20 – 10 – 11 – 25
-
99. Deepak has a brother Anil. Deepak is the son of Prem. Bimal is Prem's father. In terms of relationship, what is Anil of Bimal ?
- (1) Son (2) Grandson (3) Brother (4) Grandfather
-
100. Shubham hit the magic ball that can change directions on its own. It travelled 5 km towards east, turned left and travelled 20 km. Then it turned right, travelled 11 km and then 5 km Southwards. Then it further travelled 4 km towards east. How far is the ball from Shubham and in which direction it is ?
- (1) 21 km and North
(2) 12 km and South-West
(3) 12 km and South
(4) None of these
-



सूचना - (पृष्ठ 1 वरून पुढे.....)

- (8) प्रश्नपुस्तिकेमध्ये विहित केलेल्या विशिष्ट जागीच कच्चे काम (रफ वर्क) करावे. प्रश्नपुस्तिकेव्यतिरिक्त उत्तरपत्रिकेवर वा इतर कागदावर कच्चे काम केल्यास ते काँपी करण्याच्या उद्देशाने केले आहे, असे मानले जाईल व त्यानुसार उमेदवारावर शासनाने जारी केलेल्या “परीक्षांमध्ये होणाऱ्या गैरप्रकारांना प्रतिबंध करण्याबाबतचे अधिनियम-82” यातील तरतुदीनुसार कारवाई करण्यात येईल व दोषी व्यक्ती कमाल एक वर्षाच्या कारावासाच्या आणि/किंवा रुपये एक हजार रकमेच्या दंडाच्या शिक्षेस पात्र होईल.
- (9) सदर प्रश्नपुस्तिकेसाठी आयोगाने विहित केलेली वेळ संपल्यानंतर उमेदवाराला ही प्रश्नपुस्तिका स्वतःबरोबर परीक्षा कक्षाबाहेर घेऊन जाण्यास परवानगी आहे. मात्र परीक्षा कक्षाबाहेर जाण्यापूर्वी उमेदवाराने आपल्या उत्तरपत्रिकेचा भाग - 1 समवेक्षकाकडे न विसरता परत करणे आवश्यक आहे.

नमुना प्रश्न

Pick out the correct word to fill in the blank :

Q. No. 201. I Congratulate you _____ your grand success.

- (1) for (2) at
(3) on (4) about

ह्या प्रश्नाचे योग्य उत्तर “(3) on” असे आहे. त्यामुळे या प्रश्नाचे उत्तर “(3)” होईल. यास्तव खालीलप्रमाणे प्रश्न क्र. 201 समोरील उत्तर-क्रमांक “(3)” हे वर्तुळ पूर्णपणे छायांकित करून दाखविणे आवश्यक आहे.

प्र. क्र. 201. (1) (2) (3) (4)

अशा पद्धतीने प्रस्तुत प्रश्नपुस्तिकेतील प्रत्येक प्रश्नाचा तुमचा उत्तर-क्रमांक हा तुम्हाला स्वतंत्ररीत्या पुरविलेल्या उत्तरपत्रिकेवरील त्या त्या प्रश्न-क्रमांकासमोरील संबंधित वर्तुळ पूर्णपणे छायांकित करून दाखवावा. ह्याकरिता फक्त काळ्या शाईचे बॉलपेन वापरावे, पेन्सिल वा शाईचे पेन वापरू नये.

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